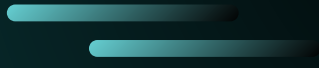




Vspaze Institute



DATA STRUCTURES & ALGORITHMS USING C++ COMPLETE CURRICULUM



Institute Overview

Vspaze Institute provides industry-focused technical training designed especially for college students and beginners.

We follow a simple, step-by-step teaching approach to help students understand concepts clearly and apply them in real-world projects.

The DSA Full Course is a complete beginner-to-job-ready program with strong practical exposure.

Detailed Course Syllabus

1. C++ Programming Foundations

- **Basic Syntax & Structure**
- **Input / Output (cin, cout)**
- **Data Types & Variables**
- **Constants • Operators**
- **Conditional Statements (if, switch)**
- **Loops (for, while, do-while)**
- **Functions**
- **Pass by Value & Reference**
- **Pointers & References**
- **Memory Basics**

2. Arrays & Strings

- **One-dimensional Arrays**
- **Two-dimensional Arrays**
- **Character Arrays**
- **Strings (std::string)**
- **String Operations**
- **Pattern Printing Problems**

3. Searching Algorithms

- Linear Search
- Binary Search
- First & Last Occurrence
- Count Occurrences
- Search in Rotated Sorted Array

4. Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Counting Sort
- Time & Space Complexity Analysis

5. Recursion & Backtracking

- Recursion Basics
- Recursive Tree
- Base & Recursive Cases
- Backtracking
- Subsets
- Permutations
- N-Queens (Intro Level)

6. Time & Space Complexity

- Big-O Notation
- Best, Average & Worst Case Analysis
- Complexity Comparison

7. Standard Template Library (STL)

- vector
- pair
- map & unordered_map
- set & unordered_set
- stack
- queue
- priority_queue
- Iterators
- Custom Comparators

8. Linked List

- Singly Linked List
- Doubly Linked List
- Insertion & Deletion
- Reverse Linked List
- Cycle Detection

9. Stack

- Stack Implementation (Array & STL)
- Stack Operations
- Valid Parentheses Problem
- Next Greater Element
- Infix, Prefix & Postfix Expressions

10. Queue

- Queue Implementation
- Circular Queue
- Deque
- Queue using Stac

11. Trees

- Binary Tree
- Tree Traversals (Inorder, Preorder, Postorder)
- Level Order Traversal
- Binary Search Tree (BST)
- Insertion, Search & Deletion in BS

12. Heaps

- Min Heap
- Max Heap
- Heap Operations
- Heap Sort
- Priority Queue using Heap

13. Hashing

- Hash Tables
- Frequency Counting
- Hash Maps
- Collision Handling Concepts
- Priority Queue using Heap

14. Graphs

- Graph Representation (Adjacency List & Matrix)
- Breadth First Search (BFS)
- Depth First Search (DFS)
- Connected Components
- Cycle Detection
- Shortest Path (Intro Level)

15. Greedy Algorithms

- Activity Selection
- Interval Scheduling
- Coin Change (Greedy)
- Job Sequencing

16. Dynamic Programming

- Introduction to Dynamic Programming
- Memoization
- Tabulation
- Fibonacci
- 0/1 Knapsack
- Subset Sum
- Longest Common Subsequence (LCS)
- Longest Increasing Subsequence (LIS)

17. Competitive Programming Concepts

- Problem Solving Techniques
- Mathematical Problems
- Edge Case Handling
- Constraints Analysis
- Fast Input / Output
- Debugging Techniques

What You Get

- ***Industry-Oriented & Future-Ready Curriculum***
- ***Hands-on Real-Time Projects***
- ***Internship Opportunity***
- ***Course Completion Certificate***
- ***Resume Building & Interview Preparation***
- ***Exclusive Job Portal & Career Support Access***
- ***100% Placement Assistance Support***
- ***Scholarships Available for Eligible Candidates***
- ***Flexible & Easy Payment Plans***



Vspaze



+91 99 0858 1431



info@vspaze.com



www.vspaze.com

Start Your Full Stack Developer Journey Today!

Become a Job-Ready C++ DSA Developer with practical training, real-time projects, and industry mentorship at Vspaze Institute.

Our Mission

To bridge the gap between academics and industry by providing practical, career-focused technology training.